NeuroScan

Brain tumor MRI classification with fine-tuned transfer learning

Dhwani Agarwal | Educational project, not a medical device



86.9%	0.868	0.974	1,137
test accuracy (was 80.4%)	macro F1 score	mean AUC, one vs rest	MRI images, 4 classes

The problem

Different brain tumors need different treatment, and the type is judged from MRI. This project trains a model to sort a slice into **Glioma**, **Meningioma**, **Pituitary Macroadenoma** or **Healthy**, as a decision-support demo.

The data

Multi-Class Brain Tumor MRI Dataset (Jamil and Khan, 2025, Mendeley Data V2, CC BY 4.0), split 80/20 and balanced:

- Training: 908 images (236 / 232 / 221 / 219)
- Testing: 229 images, untouched until final scoring

The method

1. **Prepare** — Resize to 224 x 224, three channels
2. **Augment** — Flip, rotate, zoom, contrast, brightness
3. **Train head** — MobileNetV2 frozen, new classifier, 8 epochs
4. **Fine-tune** — Unfreeze top layers, low learning rate, 25 epochs
5. **Evaluate** — Held-out test set: F1, AUC, confusion matrix

Baseline vs final

VGG16	80.4%
MobileNetV2	86.9%

+6.5 points of accuracy, from a lighter network, two-phase training, label smoothing and 224 px input.

F1 by class

Glioma	0.857
Healthy	0.881
Meningioma	0.811
Pituitary	0.923

Confusion matrix

True \ Pred.	Glioma	Healthy	Mening.	Pituit.
Glioma	54	0	5	0
Healthy	7	48	1	3
Mening.	6	2	43	5
Pituit.	0	0	1	54

Most errors are glioma vs meningioma. Pituitary recall is 98.2% and Healthy precision is 96.0%.

Training curves



What I built

- Google Colab notebook for the full pipeline
- Local Flask app for live predictions; images never leave the computer
- Nine-page learning website on tumors, MRI and the model

Limits and next steps

- Small dataset; validation by slice leaks, so validation reads 100% while test is 86.9%
- Next: patient-level split, EfficientNetV2, ensembling, in-browser TensorFlow.js

Dataset and citation

Jamil, N. and Khan, A. A. (2025). Multi-Class Brain Tumor MRI Dataset: Glioma, Healthy Brain, Meningioma, and Pituitary Macroadenoma. Mendeley Data, V2. doi:10.17632/82mtzd8x72.2. License CC BY 4.0.

Code

github.com/dhwaniagarwal13/Brain-tumor-detection-colab

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